Question

As shown in the figure, a certain amide can spontaneously form an intramolecular hydrogen bond:

This is a schematic diagram of a chemical structure, illustrating the equilibrium between two conformational isomers. On the left side of the diagram, labeled as A, the molecule exhibits a linear structure. From left to right, the structure consists of a nitrogen atom bonded to a hydrogen atom and a methyl group (Me), which is further connected to a carbonyl group (C=O). The carbonyl group is linked to a chain composed of four methylene units. The other end of the chain is attached to a second carbonyl group, which is bonded to a nitrogen atom bearing two methyl groups (Me). Between molecule A and the molecule B on the right, there is a bidirectional equilibrium arrow. The right side of the diagram shows the molecular conformation labeled as B, which adopts a cyclic folded form. In this conformation, the carbonyl oxygen atom of the gem-dimethyl carbamoyl group on the right forms an intramolecular interaction with the N-H bond of the N-methyl carbamoyl group on the left, represented by a dashed arrow pointing from the hydrogen atom to the oxygen atom. All methyl (Me) groups in this conformation are explicitly labeled.

The reaction is represented in SMILE notation as:

O=C(CCCCC(N(C)C)=O)NC>>O=C1CCCC/C(N(C)C)=O\[H]N1C

Next, the experimenter quantitatively determined the thermodynamic information of this reaction by measuring the apparent chemical shift values (δ_{obs}) of the amide at different temperatures. The measured δ_{obs} values at various temperatures are shown in the table below:

T/K	δ_{obs}/ppm
220	6.67
240	6.50
260	6.37
280	6.27
300	6.19

It is known that when the temperature is extremely low, δ_{obs} approaches $8.40\,ppm$, and when the temperature is extremely high, δ_{obs} tends toward $5.70\,ppm$. Please calculate the standard molar enthalpy change of the reaction (in $kJ \cdot mol^{-1}$), denoted as H, and the standard molar entropy change of the equilibrium transition (in $J \cdot mol^{-1} \cdot K^{-1}$), denoted as S. The following options provide a series of (H, S) values. Select the option that is closest to your calculation results.

A. (-5, -10)

B. (5,10)

C. (-20, -50)

D. (-30, -50)

E. (10, -40)

F. (20, -30)

G. (-5, -30)

H. (10,40)

I. None of the above options are correct

Answer

Correct Answer: G

Detailed Explanation

Given the problem, let the mole fractions of $\mathbf{A}$ and $\mathbf{B}$ be x_A and x_B ($x_A + x_B = 1$), respectively. Then, δ_{obs} can be expressed as:

$$\delta_{obs} = x_A\delta_A + x_B\delta_B$$

CHECKPOINT 1 PTS

$$\delta_{obs} = x_A\delta_A + x_B\delta_B$$

At low temperatures, hydrogen bonds are stable, and $\mathbf{B}$ is the dominant conformation. At high temperatures, hydrogen bonds break, and $\mathbf{A}$ is the dominant conformation. Thus:

$$\delta_{low} = \delta_B = 8.40\,ppm$$

$$\delta_{high} = \delta_A = 5.70\,ppm$$

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From this, the expression for the equilibrium constant can be derived:

$$K = \frac{x_B}{x_A} = \frac{\delta_{obs} - \delta_A}{\delta_B - \delta_{obs}} = \frac{\delta_{obs} - 5.70}{8.40 - \delta_{obs}}$$

CHECKPOINT 1 PTS

$K = \frac{x_B}{x_A} = \frac{\delta_{obs} - 5.70}{8.40 - \delta_{obs}}$

Calculate the equilibrium constant values under various temperature conditions:

T/K	K
220	0.5607
240	0.4211
260	0.3300
280	0.2676
300	0.2217

According to the van't Hoff equation, perform linear regression on $1/T - \ln K$ and calculate the slope $k = 764.2\,(K)$ and intercept $b = -4.050$.

CHECKPOINT 1 PTS

Perform linear regression on $1/T - \ln K$ and calculate the slope $k = 764.2\,(K)$ and intercept $b = -4.050$.

Thus, calculate:

$H = -k \cdot R = -6.35\,kJ \cdot mol^{-1}$

$S = b \cdot R = -33.7\,J \cdot mol^{-1} \cdot K^{-1}$

CHECKPOINT 1 PTS

$H = -6.35\,kJ \cdot mol^{-1},\ S = -33.7\,J \cdot mol^{-1} \cdot K^{-1}$

Therefore, the closest option is G.